

Grade 7
Unit 12
Lesson 4 Practice

Name _____
Date _____
Period _____

1. To find the volume of a cylinder, multiply the _____ of the base by the _____ of the cylinder.

☐ diameter
☐ radius
☐ height

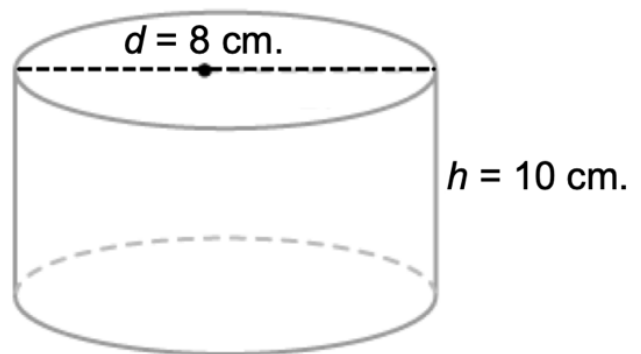
☐ area
☐ circumference

2. Which formula(s) represent the volume of a right cylinder?

- ☐ $V = r^2 \cdot h$, where r is the radius and h is the height.
☐ $V = \pi r^2 \cdot h$, where r is the radius and h is the height.
☐ $V = B \cdot h$, where B is the area of the base and h is the height.
☐ $V = C \cdot h$, where C is the circumference and h is the height.

A diagram of a right circular cylinder is shown with the diameter of 8 centimeters (cm.) and height of 10 cm for questions 3-4.

3. Calculate the area of the base. Leave in terms of π and use the appropriate units.



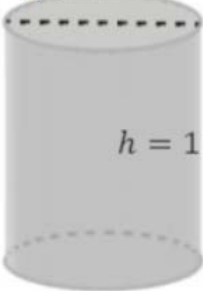
4. Calculate the volume of the cylinder. Leave in terms of π and use the appropriate units.

For questions 5-6, consider a cylinder with a radius of 12 feet (ft.) and height of 7 ft.

5. Sketch the cylinder and label its dimensions.

6. Calculate the volume of the cylinder using $\frac{22}{7}$ for π . Include appropriate units.

Four right circular cylinders are shown and specific measurements are given. The measurements include either the radius, r , diameter, d , or height, h . For each cylinder, find the volume. Use 3.14 to approximate π , round to the nearest tenth, and include appropriate units.

Cylinder	Volume
7. $h = 5$ units $r = 2$ units 	
8. $d = 8$ cm. $h = 11$ cm. 	
9. $h = 1.7$ m. $r = 3$ m. 	
10. $d = 6$ in. $h = 16$ in. 